

1.1 Informal definition of limits

Two main problems of calculus are

1. Derivative. Find the rate of change of f .
2. Integral. Find the area under a given curve.

Both are based on the concept of limit.

We say $\lim_{x \rightarrow a} f(x) = L$ to mean that $f(x)$ is “close enough” to L for any x “close enough” to a .

Example 1.1.1. *Which value is x close to when x is close to 2?*

$$\lim_{x \rightarrow 2} x = 2$$

Example 1.1.2. *Which value is 3 close to when x is close to 2?*

$$\lim_{x \rightarrow 2} 3 = 3$$

Theorem 1.1.1. *Let a and c be two real numbers. Then*

$$\lim_{x \rightarrow a} c = c, \quad \lim_{x \rightarrow a} x = a.$$

The limit $\lim_{x \rightarrow a} f(x)$ may be different from $f(a)$ as the next example shows.

Example 1.1.3.

$$f(x) = \begin{cases} x, & \text{if } x \neq 2 \\ 1, & \text{if } x = 2 \end{cases}$$

Which value is $f(x)$ close to when x is close to (but not equal to) 2?

Informal definition of left and right limits

If $f(x)$ is close to L when $x < a$ and x is close enough to a then we say

$$\lim_{x \rightarrow a^-} f(x) = L$$

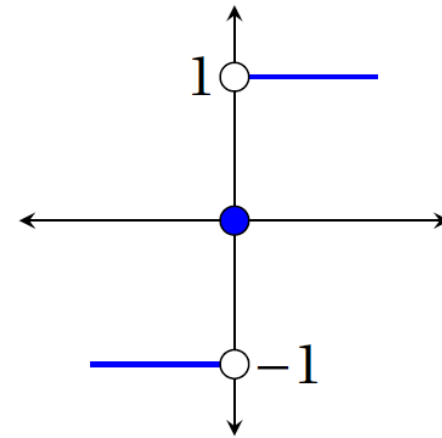
This is called the *left limit* of f at $x = a$.

Similarly we can define the right limit.

Theorem 1.1.2. $\lim_{x \rightarrow a} f(x) = L$ if and only if both $\lim_{x \rightarrow a^-} f(x) = L$ and $\lim_{x \rightarrow a^+} f(x) = L$.

Example 1.1.4. Find the left and right limits of the signum function

$$f(x) = \begin{cases} -1 & \text{for } x < 0 \\ 0 & \text{for } x = 0 \\ 1 & \text{for } x > 0 \end{cases}$$



Properties of Limits

Theorem 1.1.3. *Suppose*

$$\lim_{x \rightarrow a} f(x) = L, \quad \lim_{x \rightarrow a} g(x) = M.$$

Then

$$\lim_{x \rightarrow a} (f(x) + g(x)) = L + M,$$

$$\lim_{x \rightarrow a} (f(x) - g(x)) = L - M,$$

$$\lim_{x \rightarrow a} (f(x) \cdot g(x)) = L \cdot M$$

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad \text{if } M \neq 0$$

$$\lim_{x \rightarrow a} [f(x)]^n = L^n, \quad n = \text{positive integer}$$

$$\lim_{x \rightarrow a} [f(x)]^{1/n} = L^{1/n}, \quad n = \text{positive integer and } L > 0 \text{ if } n = \text{even}$$

Example 1.1.5. Find $\lim_{x \rightarrow 2} x^2 + 1$ and $\lim_{x \rightarrow 2} \frac{x^2 + 1}{6 - x}$.

Theorem 1.1.4. *If $P(x)$ is a polynomial then,*

$$\lim_{x \rightarrow a} P(x) = P(a)$$

If $Q(x)$ is another polynomial with $Q(a) \neq 0$ then

$$\lim_{x \rightarrow a} \frac{P(x)}{Q(x)} = \frac{P(a)}{Q(a)}.$$

Theorem 1.1.5. Suppose that $f(x) \leq g(x) \leq h(x)$ and $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$. Then $\lim_{x \rightarrow a} g(x) = L$.

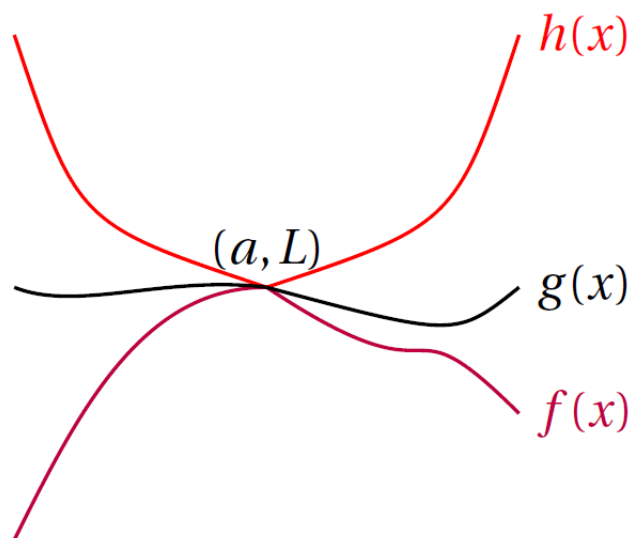


Figure 1.1: The Squeeze Theorem.

Example 1.1.6. *If $-x^2 \leq g(x) \leq x^2$ for $-1 \leq x \leq 1$, find $\lim_{x \rightarrow 0} g(x)$.*

Example 1.1.7. *Show that if $\lim_{x \rightarrow a} |f(x)| = 0$ then $\lim_{x \rightarrow a} f(x) = 0$.*

Example 1.1.8. *Let*

$$f(x) = \frac{|x - 2|}{x^2 + x - 6}.$$

Find $\lim_{x \rightarrow 2+} f(x)$, $\lim_{x \rightarrow 2-} f(x)$. Does $\lim_{x \rightarrow 2} f(x)$ exist?

Example 1.1.9.

$$\lim_{x \rightarrow -2} \frac{x^2 + x - 2}{x^2 + 5x + 6},$$

Exercises.

$$1. \lim_{x \rightarrow 5} \frac{\frac{1}{x} - \frac{1}{5}}{x - 5}.$$

$$2. \lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x^2 - 16},$$

$$3. \lim_{x \rightarrow -2} \frac{x^2 + 2x}{x^2 - 4}.$$

$$4. \lim_{h \rightarrow 0} \frac{\sqrt{4+h} - 2}{h}.$$

$$5. \lim_{t \rightarrow 0} \frac{t}{\sqrt{4+t} - \sqrt{4-t}}.$$

6. $\lim_{x \rightarrow -1} \frac{x^3 + 1}{x + 1}.$

7. $\lim_{x \rightarrow 0} \frac{|3x - 1| - |3x + 1|}{x}.$

8. $\lim_{x \rightarrow -2^-} \frac{x^2 - 4}{|x + 2|}.$

9. $\lim_{y \rightarrow 1} \frac{y - 4\sqrt{y} + 3}{y^2 - 1}.$

10. $\lim_{x \rightarrow 2} \frac{\sqrt{4 - 4x + x^2}}{x - 2}.$

1.2 Limits at Infinity and Infinite Limits

Limits at Infinity

Definition 1.2.1. We will say that $\lim_{x \rightarrow \infty} f(x) = L$ if $f(x)$ is “close enough” to L whenever $x > 0$ is “large enough”.

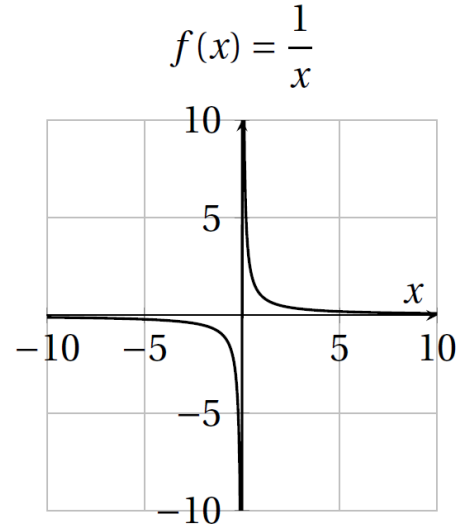
Similarly we define $\lim_{x \rightarrow -\infty} f(x) = L$ if $f(x)$ is “close enough” to L whenever $x < 0$ is “large enough”.

If either $\lim_{x \rightarrow \infty} f(x) = L$ or $\lim_{x \rightarrow -\infty} f(x) = L$, we say that the line $y = L$ is an **horizontal asymptote** of the graph of f .

Example 1.2.1. *Argue that*

$$\lim_{x \rightarrow \infty} 1/x = \lim_{x \rightarrow -\infty} 1/x = 0.$$

by making a table of values of x and $1/x$.



Recall that for ordinary limits, limit of product of functions is a product of limits of functions. Same is also true for limits at infinity. Hence

$$\lim_{x \rightarrow \infty} \frac{1}{x^2} = \lim_{x \rightarrow \infty} \frac{1}{x} \cdot \lim_{x \rightarrow \infty} \frac{1}{x} = 0 \times 0 = 0.$$

Similarly

$$\lim_{x \rightarrow -\infty} \frac{1}{x^2} = 0$$

Finally, for any positive integer n

$$\lim_{x \rightarrow \infty} \frac{1}{x^n} = \lim_{x \rightarrow -\infty} \frac{1}{x^n} = 0.$$

Example 1.2.2. Let $f(x) = \frac{x}{\sqrt{x^2 + 1}}$. Find $\lim_{x \rightarrow \infty} f(x)$, $\lim_{x \rightarrow -\infty} f(x)$.

Limits of Rational Functions at Infinity

Recall that a rational function is a ratio of two polynomials.

Strategy. To find limits of rational functions at infinity, divide by the highest power of x appearing in the *denominator*.

Example 1.2.3.

$$\lim_{x \rightarrow \pm\infty} \frac{2x^2 - x + 3}{3x^2 + 5} :$$

Example 1.2.4.

$$\lim_{x \rightarrow \pm\infty} \frac{x - 5}{2x^2 + 4x + 1} :$$

Theorem 1.2.1. *Let $P(x) = a_px^p + a_{p-1}x^{p-1} + \cdots + a_0$ be a polynomial of degree p and $Q(x) = b_qx^q + \cdots + b_0$ be a polynomial of degree q . If $p = q$, then*

$$\lim_{x \rightarrow \pm\infty} \frac{P(x)}{Q(x)} = \frac{a_p}{q_p},$$

If $p < q$, then

$$\lim_{x \rightarrow \pm\infty} \frac{P(x)}{Q(x)} = 0,$$

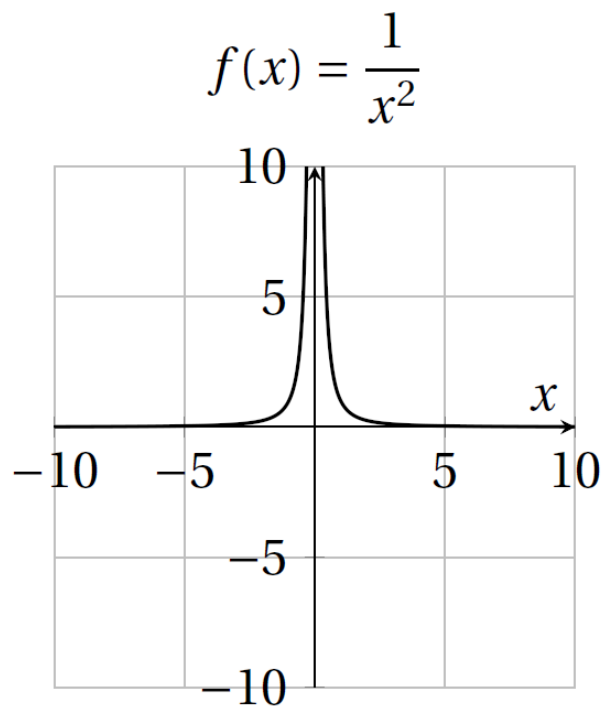
Example 1.2.5.

$$\lim_{x \rightarrow \infty} \sqrt{x^2 + x} - x :$$

Infinite Limits

Example 1.2.6. *The values of $\frac{1}{x^2}$ gets larger and larger as x approaches to 0. Thus $\lim_{x \rightarrow 0} \frac{1}{x^2}$ does not exist. Although the limit does not exist, it is useful to state why it does not exist by writing*

$$\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty.$$



Example 1.2.7.

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty.$$

$$\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty.$$

$$\lim_{x \rightarrow 0} \frac{1}{x} \text{ does not exist.}$$

Example 1.2.8.

$$\lim_{x \rightarrow -\infty} \sqrt{x^2 + x} - x$$

Solution. Both $-x$ and $\sqrt{x^2 + x}$ grow large as $x \rightarrow -\infty$. So the limit is ∞ .

Behaviour of Polynomials at Infinity

Example 1.2.9.

$$\lim_{x \rightarrow \infty} 4x^3 - 2x + 1 = \lim_{x \rightarrow \infty} 4x^3 = \infty.$$

$$\lim_{x \rightarrow -\infty} -3x^5 + x^3 + 1 = \lim_{x \rightarrow -\infty} -3x^5 = \infty.$$

In general,

Theorem 1.2.2. *If $P(x) = a_n x^n + \cdots + a_0$ is a polynomial then*

$$\lim_{x \rightarrow \pm\infty} P(x) = \lim_{x \rightarrow \pm\infty} a_n x^n.$$

If $Q(x) = b_m x^m + \cdots + b_0$ is also a polynomial then

$$\lim_{x \rightarrow \pm\infty} \frac{P(x)}{Q(x)} = \lim_{x \rightarrow \pm\infty} \frac{a_n x^n}{b_m x^m}$$

Example 1.2.10.

$$\lim_{x \rightarrow \infty} \frac{x^3 + 1}{x^2 - 2x}$$

Example 1.2.11. 1. $\lim_{x \rightarrow 2} \frac{(x-2)^2}{x^2 - 4} = 0$

2. $\lim_{x \rightarrow 2^+} \frac{x-3}{x^2 - 4} = -\infty$

3. $\lim_{x \rightarrow 2^-} \frac{x-3}{x^2 - 4} = \infty$

4. $\lim_{x \rightarrow 2} \frac{x-3}{x^2 - 4}$ *does not exist.*

5. $\lim_{x \rightarrow \infty} \frac{2x-1}{\sqrt{3x^2 + x + 1}},$

1.3 Continuity

Let $f(x) = \sqrt{4 - x^2}$. Domain of f is $[-2, 2]$.

- $x = -2$ is the left end point of $\text{Dom}(f)$.
- $x = 2$ is the right end point of $\text{Dom}(f)$.
- Any x with $-2 < x < 2$ is called an interior point of $\text{Dom}(f)$.

Definition 1.3.1. A function f is **continuous** at an interior point c of its domain if

$$\lim_{x \rightarrow c} f(x) = f(c)$$

f is continuous at its left endpoint c if

$$\lim_{x \rightarrow c+} f(x) = f(c)$$

f is continuous at its right endpoint c if

$$\lim_{x \rightarrow c-} f(x) = f(c)$$

Theorem 1.3.1. *A function f is **continuous** at an interior point c of its domain if and only if*

$$\lim_{h \rightarrow 0} f(c+h) = f(c)$$

f is continuous at its left endpoint c if

$$\lim_{h \rightarrow 0+} f(c+h) = f(c)$$

f is continuous at its right endpoint c if

$$\lim_{h \rightarrow 0-} f(c+h) = f(c)$$

Proof. Let $h = x - c$. Then $x \rightarrow c$ if and only if $h \rightarrow 0$. So $\lim_{h \rightarrow 0} f(c+h) = f(c)$ is the same as $\lim_{h \rightarrow 0} f(c+h) = f(c)$. □

Note that f is discontinuous at c if

- i) either $\lim_{x \rightarrow c} f(x)$ does not exist.
- ii) or $\lim_{x \rightarrow c} f(x)$ exists but is not equal to $f(c)$.

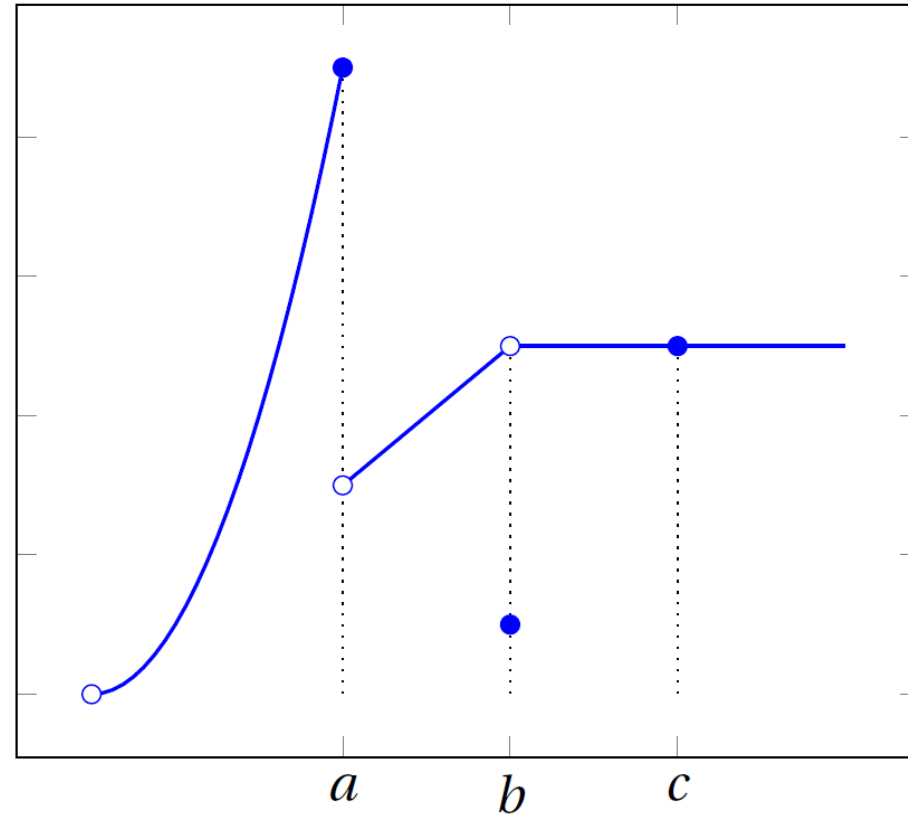
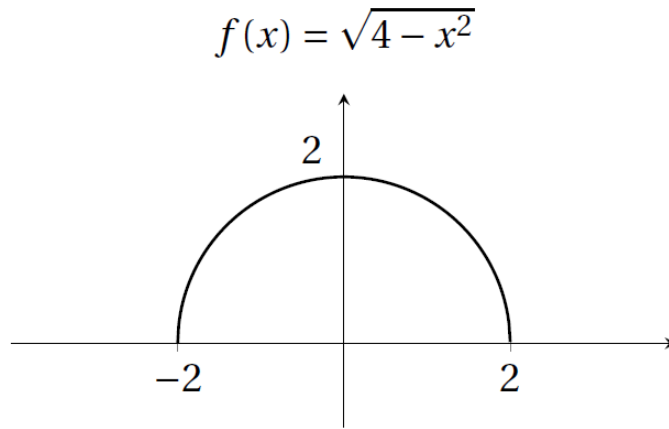


Figure 1.2: f is discontinuous at a because of (ii) and discontinuous at b because of (i). f is continuous at c .

Example 1.3.1. $f(x) = \sqrt{4 - x^2}$ is continuous at every point of its domain. So it is a continuous function.



According to this definition $f(x) = \frac{1}{x}$ is also continuous!!! 0 is not in domain of f . So we say f is undefined rather than discontinuous at 0.

Theorem 1.3.2. *If f and g are continuous at c then*

- $f + g, f - g, fg$, are continuous at c ,
- if k is constant then kf is continuous at c ,
- $\frac{f}{g}$ is continuous at c provided that $g(c) \neq 0$.
- $f^{1/n}$ continuous at c provided that $f(c) > 0$ if n is even.

Proof. Let's prove that if f and g are continuous at c then so is $f + g$. If f and g are continuous at c then

$$\lim_{x \rightarrow c} f(x) = f(c), \quad \lim_{x \rightarrow c} g(x) = g(c),$$

By the limit rule,

$$\lim_{x \rightarrow c} (f(x) + g(x)) = \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x) = f(c) + g(c).$$

The other proofs are similar.

Composites of continuous functions are continuous

If g is continuous at c and f is continuous at $g(c)$ then $f \circ g$ is continuous at c . In other words,

$$\lim_{x \rightarrow c} f(g(x)) = f(\lim_{x \rightarrow c} g(x)) = f(g(c)).$$

Example 1.3.2. Find m so that

$$g(x) = \begin{cases} x - m, & \text{if } x < 3, \\ 1 - mx, & \text{if } x \geq 3 \end{cases}$$

is continuous for all x .

Continuity of Trigonometric Functions

Theorem 1.3.3. $\sin x$ and $\cos x$ are continuous at $x = 0$, i.e.

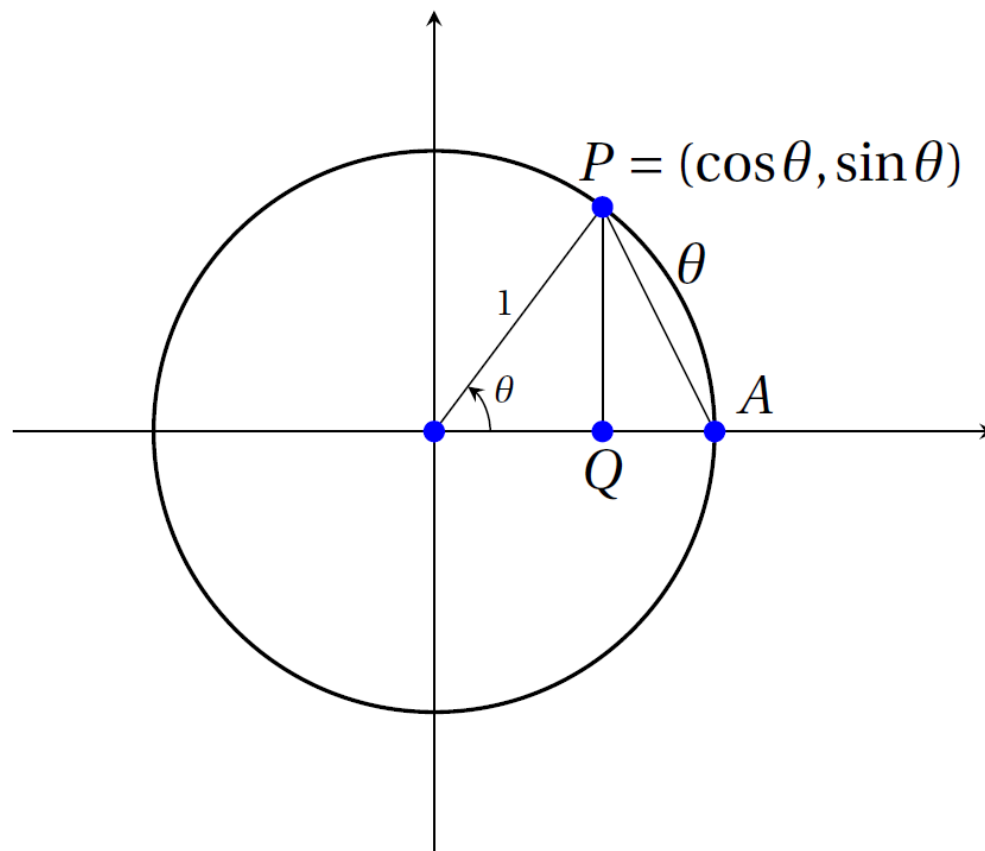
$$\lim_{x \rightarrow 0} \sin x = \sin 0 = 0, \quad \lim_{x \rightarrow 0} \cos x = \cos 0 = 1.$$

Proof.

$$|1 - \cos \theta| = |AQ| \leq |AP| \leq \theta,$$

$$|\sin \theta| = |PQ| \leq |AP| \leq \theta$$

In other words, $-\theta \leq \sin \theta \leq \theta$ and using the squeeze theorem we get $\lim_{\theta \rightarrow 0} \sin \theta = 0$. Similarly, we get $\lim_{\theta \rightarrow 0} 1 - \cos \theta = 0$ or $\lim_{\theta \rightarrow 0} \cos \theta = 1$.



Theorem 1.3.4. $\sin x$ and $\cos x$ are continuous for all x .

Proof. By Theorem 1.3.1, we need to prove $\lim_{h \rightarrow 0} \sin(x + h) = \sin x$ for any x .

$$\lim_{h \rightarrow 0} \sin(x + h) = \lim_{h \rightarrow 0} \sin x \cos h + \cos x \sin h = \sin x \lim_{h \rightarrow 0} \cos h + \cos x \lim_{h \rightarrow 0} \sin h = \sin x.$$

Extreme Value Theorem

Theorem 1.3.5. *If f is continuous on the closed interval $[a, b]$ then there exist numbers p and q in the interval $[a, b]$ s.t.*

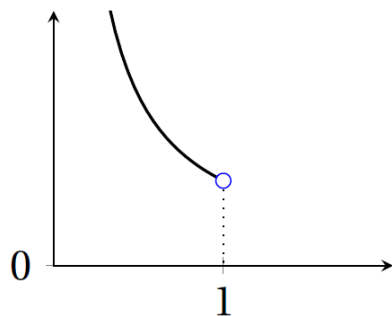
$$f(p) \leq f(x) \leq f(q)$$

*for all x in $[a, b]$. $f(p)$ is called the **absolute minimum value** and $f(q)$ is called the **absolute maximum value**.*

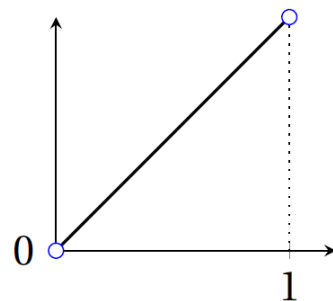
Extreme value theorem is an existence theorem. It only guarantees the existence of p and q but does not tell how to actually find them.

We say a function f is **bounded** if there exists M and N such that $M \leq f(x) \leq N$ for all x in the domain of f . Extreme value theorem says that continuous functions on closed intervals must be bounded.

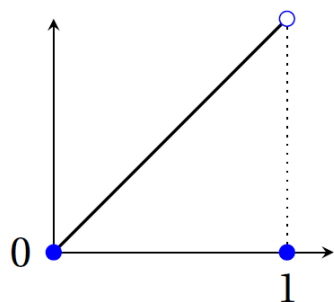
Example 1.3.3. *The conclusions of the theorem may fail if the function f is not continuous or the interval is not closed.*



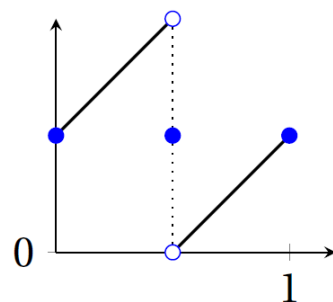
(a) The function $f(x) = 1/x$ on the open interval $(0, 1)$ is continuous but unbounded and has no minimum and no maximum.



(b) The function $f(x) = x$ on $(0, 1)$ is discontinuous, bounded and has no minimum and no maximum.



(a) This function is defined on the closed interval $[0, 1]$, discontinuous, has a minimum but no maximum.



(b) This function is defined on the closed interval $[0, 1]$, discontinuous, bounded, has no minimum and no maximum.

Intermediate Value Theorem

Theorem 1.3.6 (Intermediate Value Theorem). *If f is continuous on $[a, b]$ and if s is between $f(a)$ and $f(b)$ then there exists c in $[a, b]$ s.t. $f(c) = s$.*

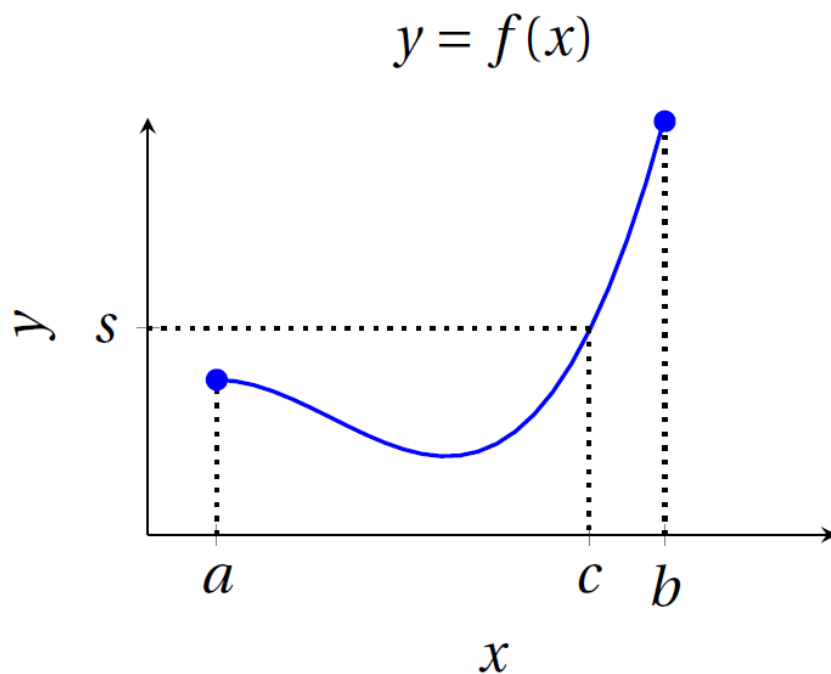


Figure 1.5: Illustration of the intermediate value theorem.

Example 1.3.4. *If a child grows from 1 m to 1.5 m between the ages of two and six years, then, at some time between two and six years of age, the child's height must have been 1.23 m.*

In particular, a continuous function on a closed interval takes every value between its minimum m and maximum M . Hence its range is a closed interval $[m, M]$.

Example 1.3.5. *Show that the equation $x^3 - x - 1 = 0$ has a solution in the interval $[1, 2]$.*

Solution. $f(x) = x^3 - x - 1$ is a polynomial and hence continuous. $f(1) = -1$ and $f(2) = 5$. Since 0 lies between -1 and 5 , the intermediate value theorem assures us that there must be a number c in $[1, 2]$ such that $f(c) = 0$.

1.5 Review Problems

Example 1.5.1. *Evaluate the limits if they exist. If they do not exist, state whether they are ∞ , $-\infty$ or just does not exist.*

1. $\lim_{x \rightarrow 2} \frac{x^2 + 1}{1 - x^2},$

2. $\lim_{x \rightarrow 1} \frac{x^2}{1 - x^2},$

3. $\lim_{x \rightarrow \infty} \frac{\cos x}{x},$ (Hint: Use Sandwich Theorem)

4. $\lim_{x \rightarrow -\infty} \frac{2x^3 + 2x - 1}{-3x^3 + x^2},$

5. $\lim_{x \rightarrow -\infty} x + \sqrt{x^2 - 4x + 1},$

6. $\lim_{x \rightarrow 0} \frac{x}{|x-1| - |x+1|}.$

7. $\lim_{x \rightarrow 5} \frac{x-5}{x^2-25}$

8. $\lim_{x \rightarrow -5} \frac{x^2+3x-10}{x+5}$

9. $\lim_{x \rightarrow 1} \frac{x^{-1}-1}{x-1}$

10. $\lim_{u \rightarrow 1} \frac{u^4 - 1}{u^3 - 1}$

11. $\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9}$

12. $\lim_{x \rightarrow 0} \frac{\frac{1}{x-1} + \frac{1}{x+1}}{x}$

13. $\lim_{v \rightarrow 2} \frac{v^3 - 8}{v^4 - 16}$

14. If $2 - x^2 \leq g(x) \leq 2 \cos x$, for all x , find $\lim_{x \rightarrow 0} g(x)$

15. If $\lim_{x \rightarrow 2} \frac{f(x) - 5}{x - 2} = 3$, find $\lim_{x \rightarrow 2} f(x)$

16. If $\lim_{x \rightarrow 2} \frac{f(x) - 5}{x - 2} = 4$, find $\lim_{x \rightarrow 2} f(x)$

17. $\lim_{h \rightarrow 0+} \frac{\sqrt{h^2 + 4h + 5} - \sqrt{5}}{h}$

18. $\lim_{x \rightarrow -2-} (x + 3) \frac{|x + 2|}{x + 2}$

19. *Define $g(3)$ in a way that extends $g(x) = (x^2 - 9)/(x - 3)$ to be a continuous at $x = 3$.*

20. *For what value of b is*

$$g(x) = \begin{cases} ax + 2b, & x \leq 0 \\ x^2 + 3a - b, & 0 < x \leq 2 \\ 3x - 5, & x > 2 \end{cases}$$

continuous at every x ?